Individual Assignment 2

Pilgrim Bank (B): Customer Retention Prediction Using Nonlinear Machine Learning Models

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Abstract

Customer retention plays a critical role in long-term profitability for banks, as retaining existing customers is typically far more cost-effective than acquiring new ones. In the Pilgrim Bank (B) case, the primary strategic question is whether the bank can meaningfully predict customer retention using available behavioral, demographic, and channel-usage attributes. In this study, my machine learning analysis focuses on predicting customer retention for Pilgrim Bank. Building upon the initial baseline logistic regression model, I extend the analysis to include five nonlinear models: Decision Tree, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Random Forest, and XGBoost. To evaluate the models, I consider the measures such as accuracy, precision, recall, F1 score, and the Area Under the Curve (AUC). The AUC, F1, and confusion matrix evaluations confirm that these flexible models better capture nonlinear patterns and interactions in customer behavior. The study concludes that nonlinear machine learning methods offer superior predictive performance and more robust insight into customer retention drivers, providing Pilgrim Bank with a stronger toolkit for proactive churn-reduction strategies.

1.Introduction

Customer retention is a key determinant of long-term financial performance in the banking sector, particularly as market competition and declining switching costs increase the likelihood of customer defection. In this study, retention between 1999 and 2000 is inferred from the presence or absence of customer records across years, using demographic, behavioral, and financial attributes.

While logistic regression is commonly used due to its transparency, its assumptions, linear relationships, additive effects, and constant marginal influence do not reflect the complex, nonlinear nature of customer behavior. Retention decisions often involve thresholds, interactions, and nonlinear patterns (e.g., distinct behaviors among new, mid-tenure, and long-tenure customers), which linear models cannot capture effectively.

2. Objectives

The goal in this study is to enhance predictive accuracy and gain a deeper understanding of the drivers of customer loyalty. Hence the steps are: First, to develop a complete machine learning pipeline for retention prediction. Then, implement five nonlinear models and conduct a comparison study between the baseline regression model and the nonlinear models. Provide managerial implications, recommendations, and conclusions.

3. Data Preparation and Feature Engineering

In this section, I outline the steps involved in data preparation for predictive modeling. The Pilgrim Bank (B) dataset includes customer demographics, financial attributes, online behavior, and bill-pay usage. Several preprocessing are given below:

3.1 Handling missing values and defining Retention

Missing values in age and income are imputed to preserve sample size and avoid bias. The target variable, *Retention*, is constructed by identifying whether customers appeared in the 2000 dataset, distinguishing retained customers from those who churned.

3.2 Encoding and scaling

The district was one-hot encoded to capture categorical geographic information. Numerical variables, including age, income, tenure, and bill-pay activity, are standardized using *StandardScaler*, a necessary step for distance-based and gradient-based models such as SVM and XGBoost.

3.3 Train–test split and unified pipeline

The data were split into training and testing sets using an 80/20 partition. All preprocessing, including imputation, encoding, and scaling, is consolidated within a unified *ColumnTransformer* pipeline to ensure reproducibility and prevent data leakage during model training and validation.

These steps provide each nonlinear model with clean, properly scaled, and consistently formatted features, supporting robust and reliable retention predictions.

4 Nonlinear Machine Learning Models

To more effectively capture the complex behavioral and demographic patterns associated with customer retention at Pilgrim Bank, five nonlinear machine learning models were developed and evaluated (See Appendix for detailed charts and results). These models are selected because they can represent interactions, thresholds, and nonlinear relationships that logistic regression cannot adequately capture. The following five nonlinear machine learning models are implemented:

I. **Decision Tree:** Useful for interpretability, capturing hierarchical splits.
II. **K-Nearest Neighbors:** A distance-based classifier sensitive to scaling and feature distributions.
III. **Support Vector Machine:** Capable of modeling nonlinear boundaries.
IV. **Random Forest:** An ensemble of decision trees offering robustness and reduced overfitting.
V. **XGBoost:** A gradient boosting model known for high performance on structured data.

All models are trained using **GridSearchCV** with **5-fold cross-validation** to systematically explore hyperparameters and prevent overfitting.

5. Model Performance Comparison

Table 1 provides the full numerical comparison across all models using test-set performance metrics.

Model	Accuracy	Precision	Recall	F1 Score	AUC
Random Forest	0.763	0.924	0.780	0.846	0.767
XGBoost	0.838	0.849	0.981	0.910	0.764
Decision Tree	0.742	0.929	0.749	0.829	0.763
KNN	0.832	0.856	0.961	0.906	0.750
SVM	0.686	0.922	0.681	0.783	0.732
Logistic Regression (baseline)	0.532	0.897	0.496	0.639	0.647

The expanded results show that nonlinear models outperform logistic regression across all metrics. XGBoost and Random Forest deliver the highest predictive performance, with XGBoost excelling in recall and F1-score, making it ideal for identifying at-risk customers. Logistic regression performs poorly, indicating it is unsuitable for capturing nonlinear behavior.

6. Managerial Implications and Recommendations

The nonlinear analysis highlights the predictors most strongly associated with retention, notably income, age, tenure, and digital engagement. Managers should focus on early engagement for low-tenure customers and promote online services that are correlated with higher customer retention rates. XGBoost and Random Forest models can be integrated into CRM systems for risk scoring, enabling proactive outreach to potential churners. Retention strategies should prioritize customers with medium-to-high churn probability as identified by the model.

7. Conclusion

Based on this study, it can be concluded that nonlinear machine learning models significantly outperform logistic regression in predicting customer retention at Pilgrim Bank. XGBoost demonstrates the strongest recall and a high F1-score, while Random Forest provides the most balanced overall performance. The Appendix provides a comprehensive set of diagnostic charts that support these findings.

Appendix: Figures and Visualizations

All charts, confusion matrices, and diagnostic visualizations are included in this Appendix.

Figure: Customer Retention Distribution.png

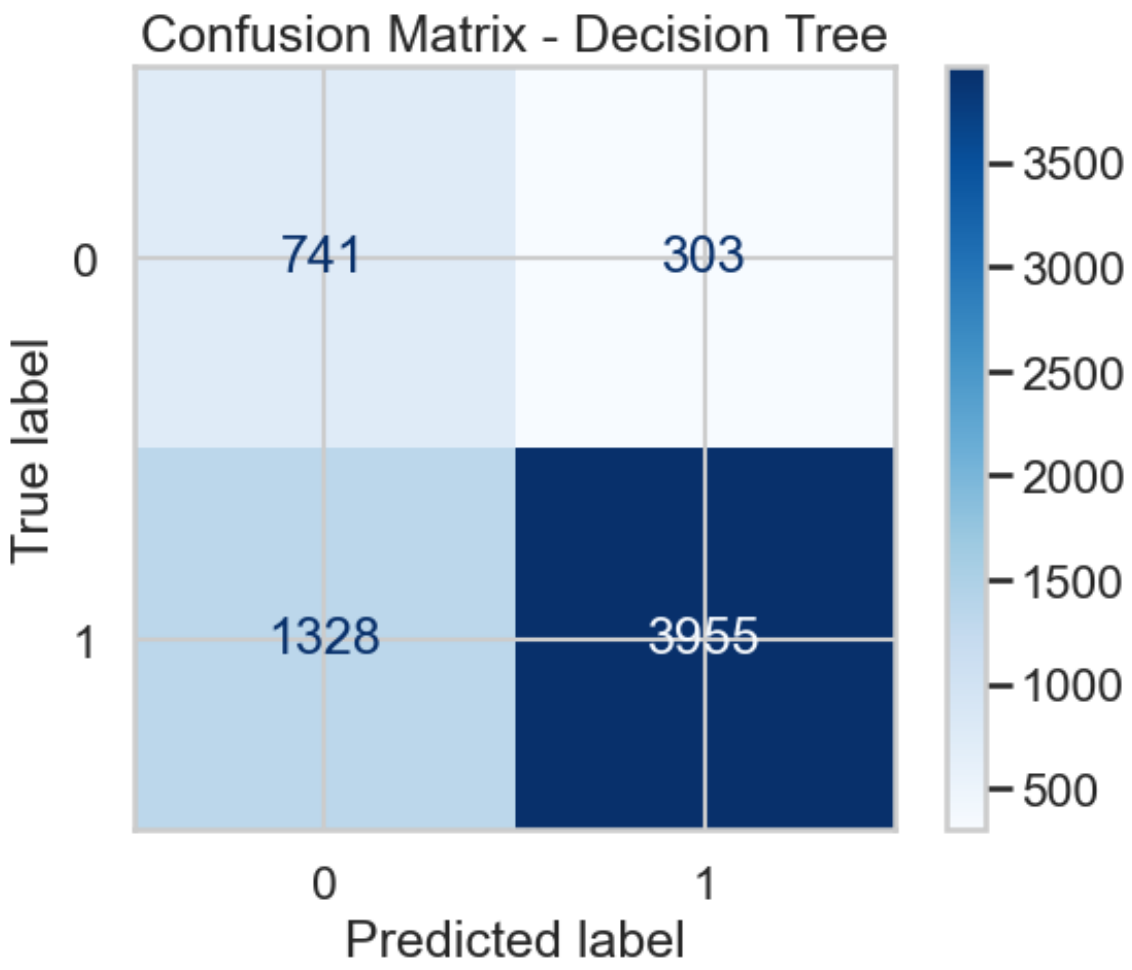

Figure: Confusion Matrix Decision Tree.png

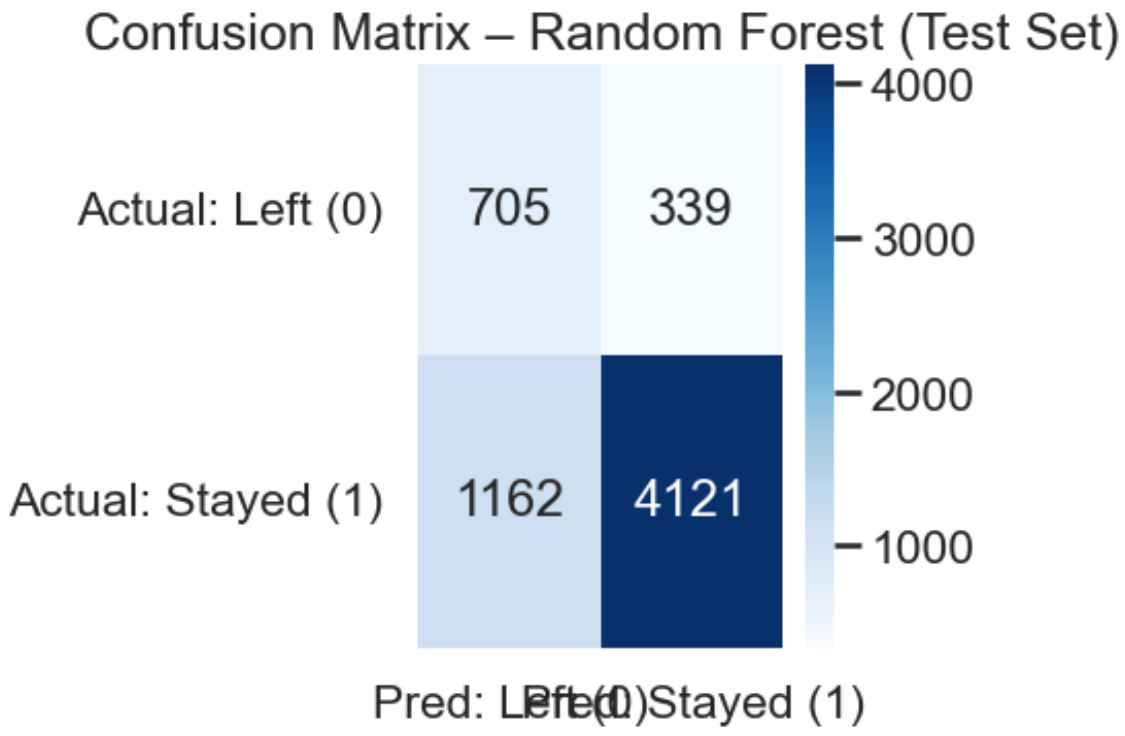

Figure: Confusion Matrix Random Forest.png

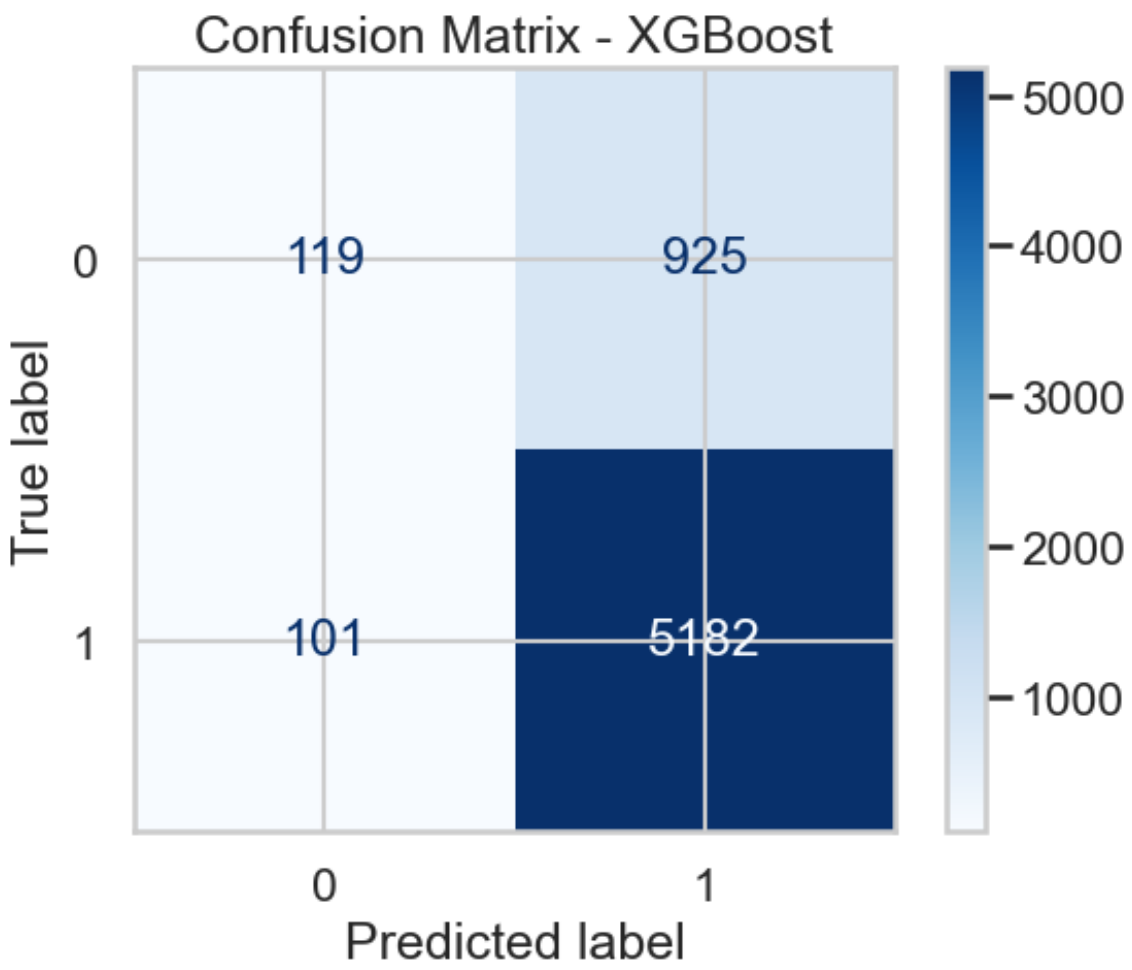

Figure: Confusion Matrix for XGBoost.png

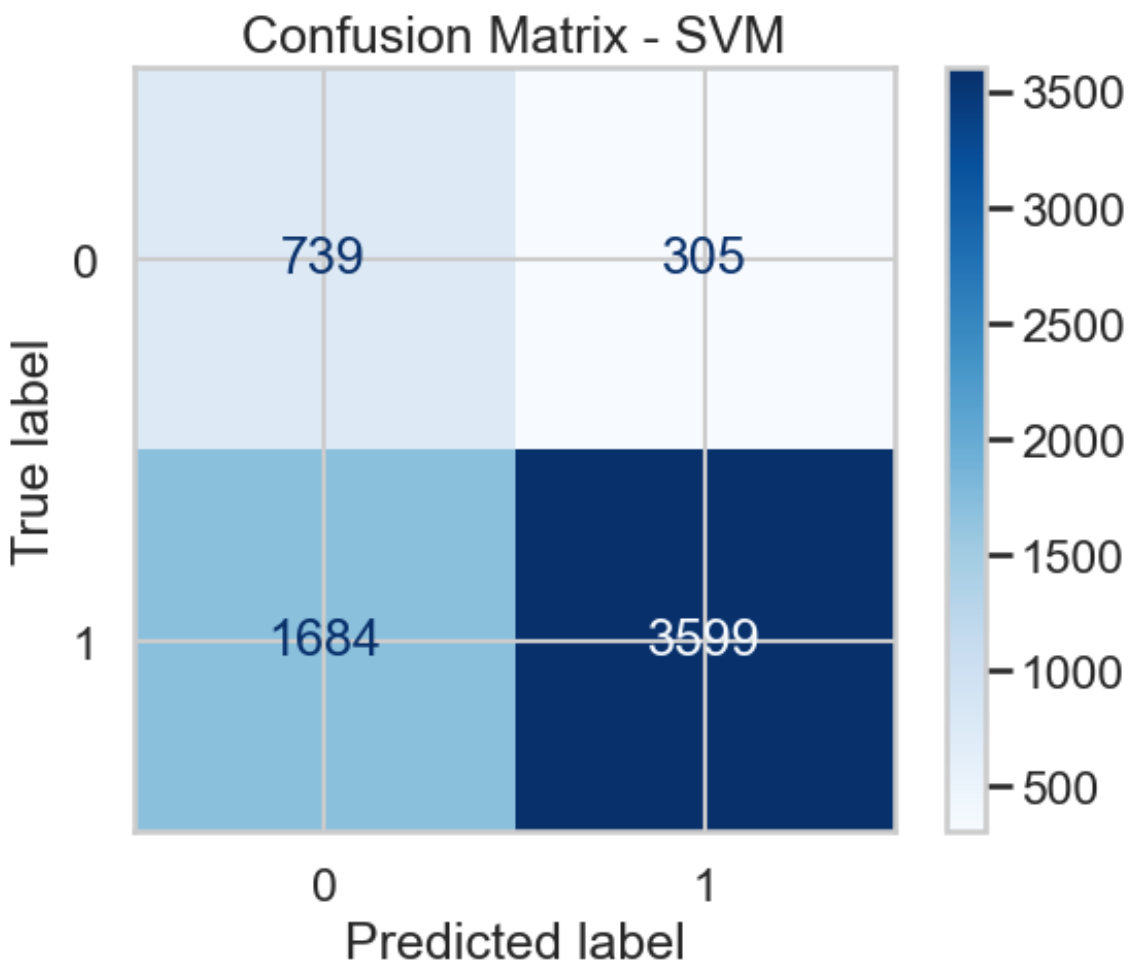

Figure: Confusion Matrix SVM.png

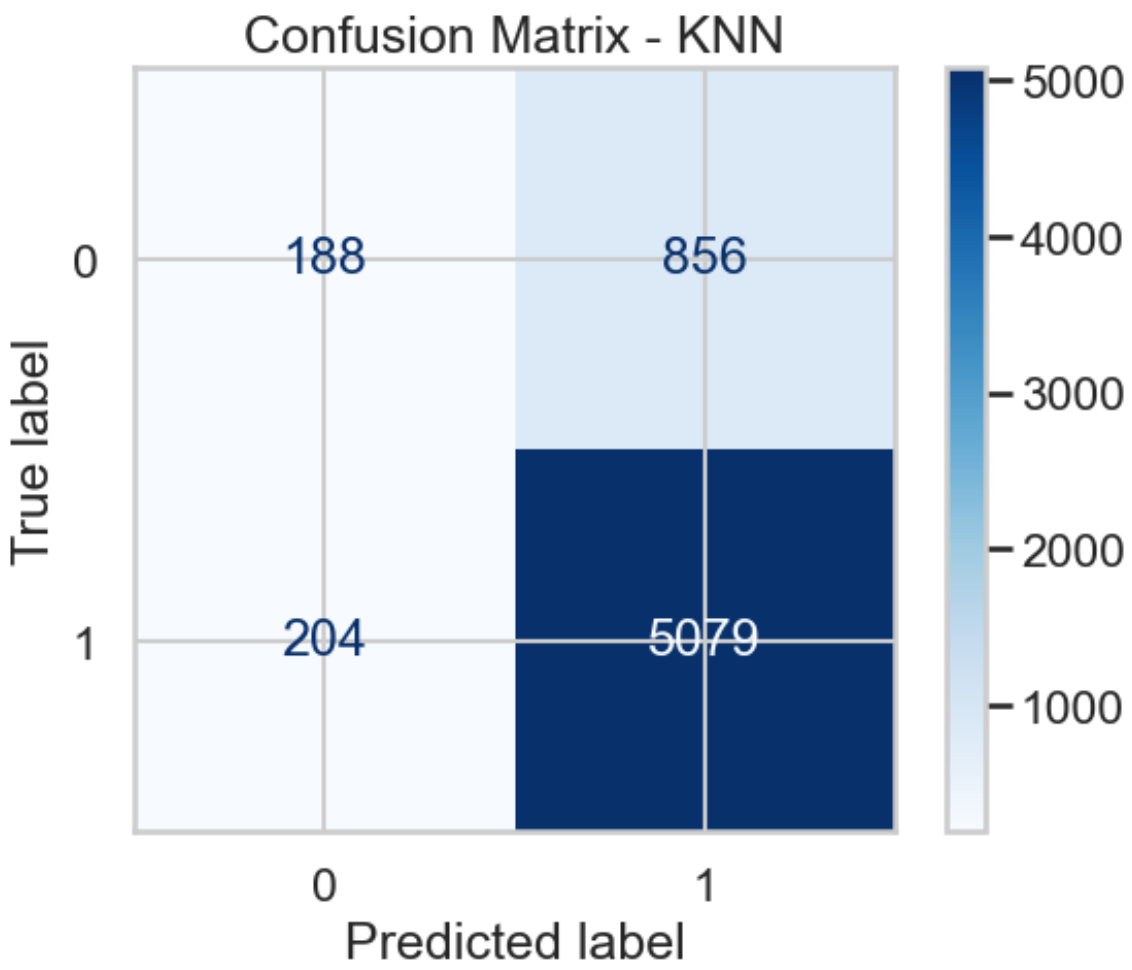

Figure: KNN Confusion Matrix.png

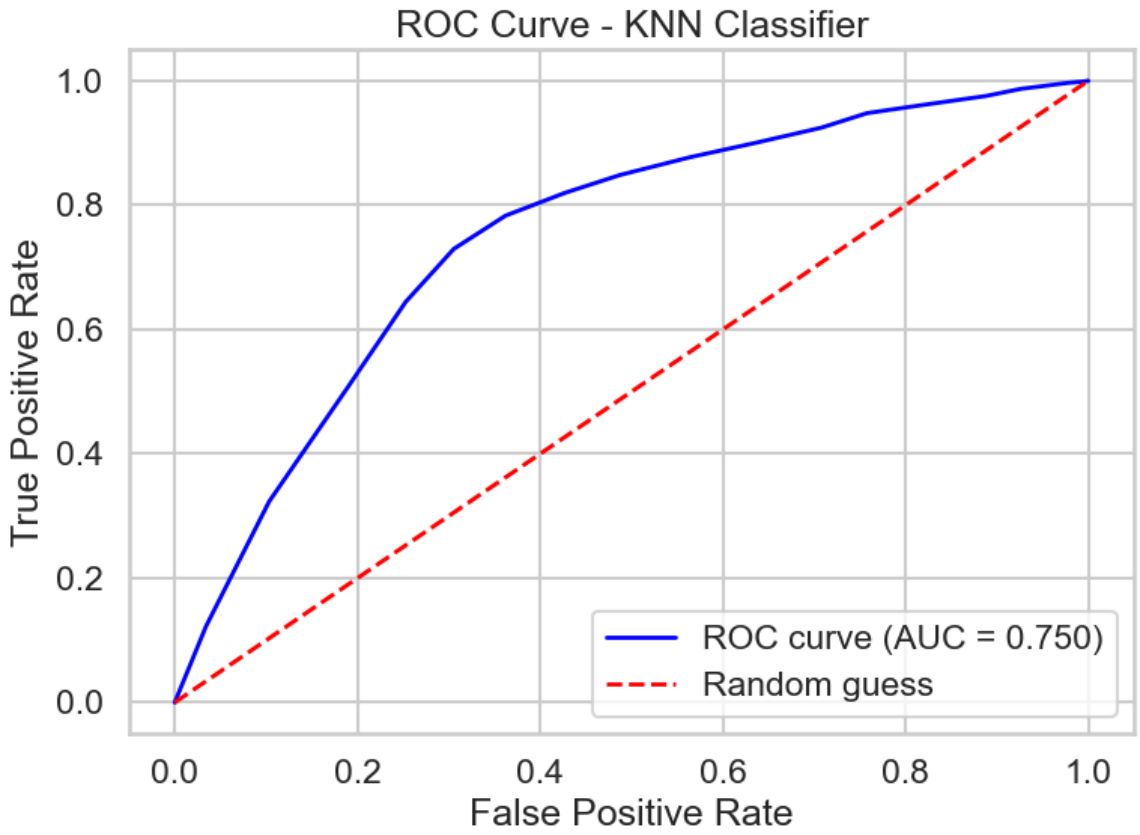

Figure: ROC KNN Classifier.png

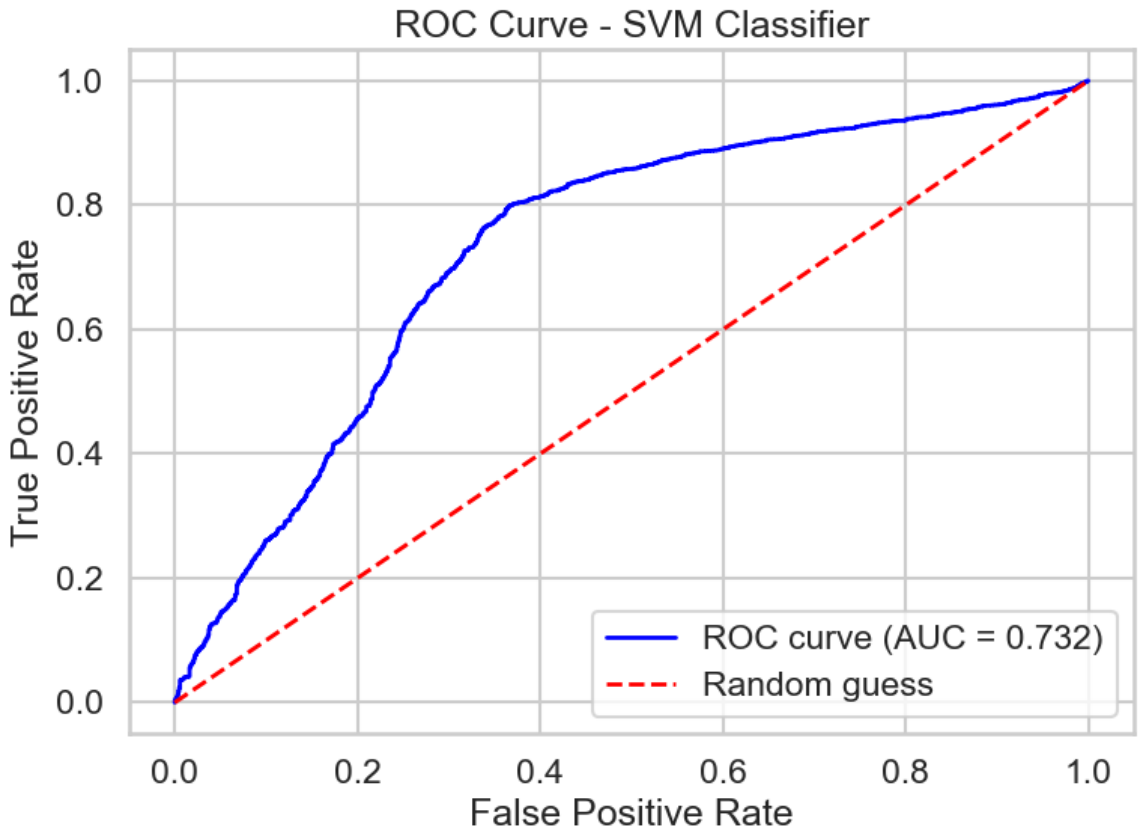

Figure: ROC SVM.png

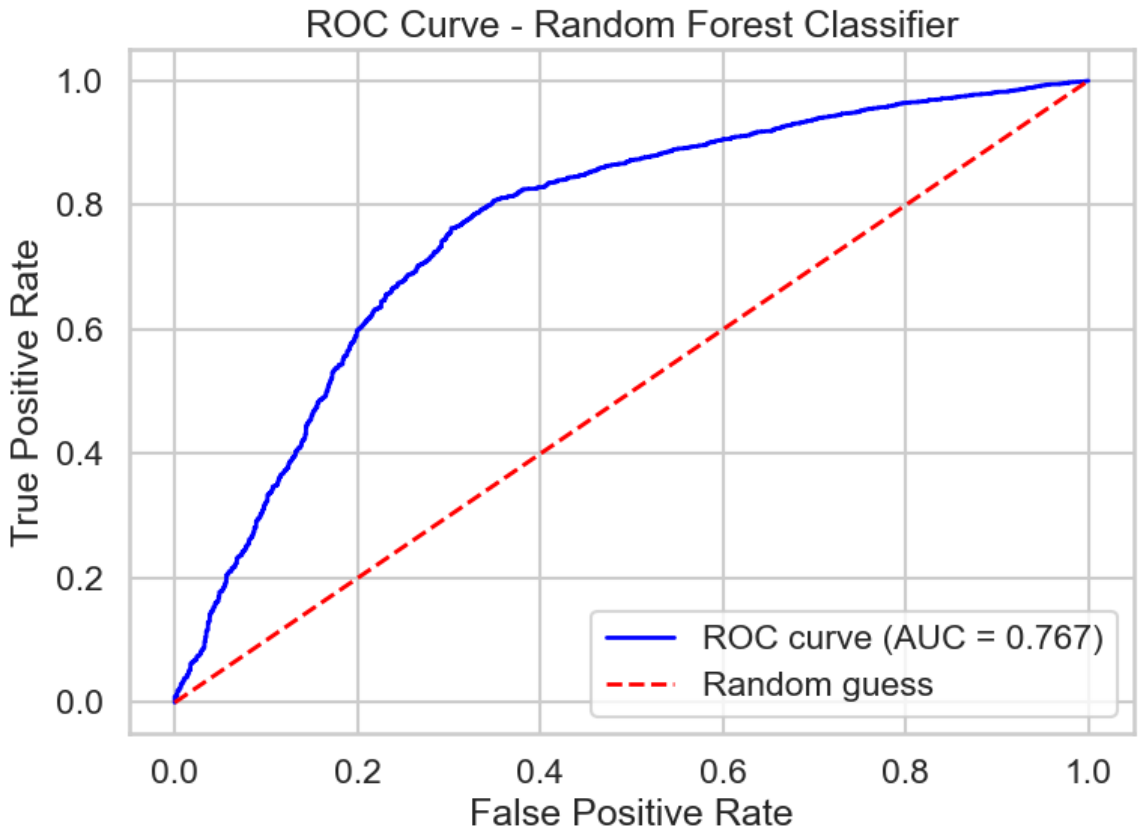

Figure: ROC-Random Forest.png

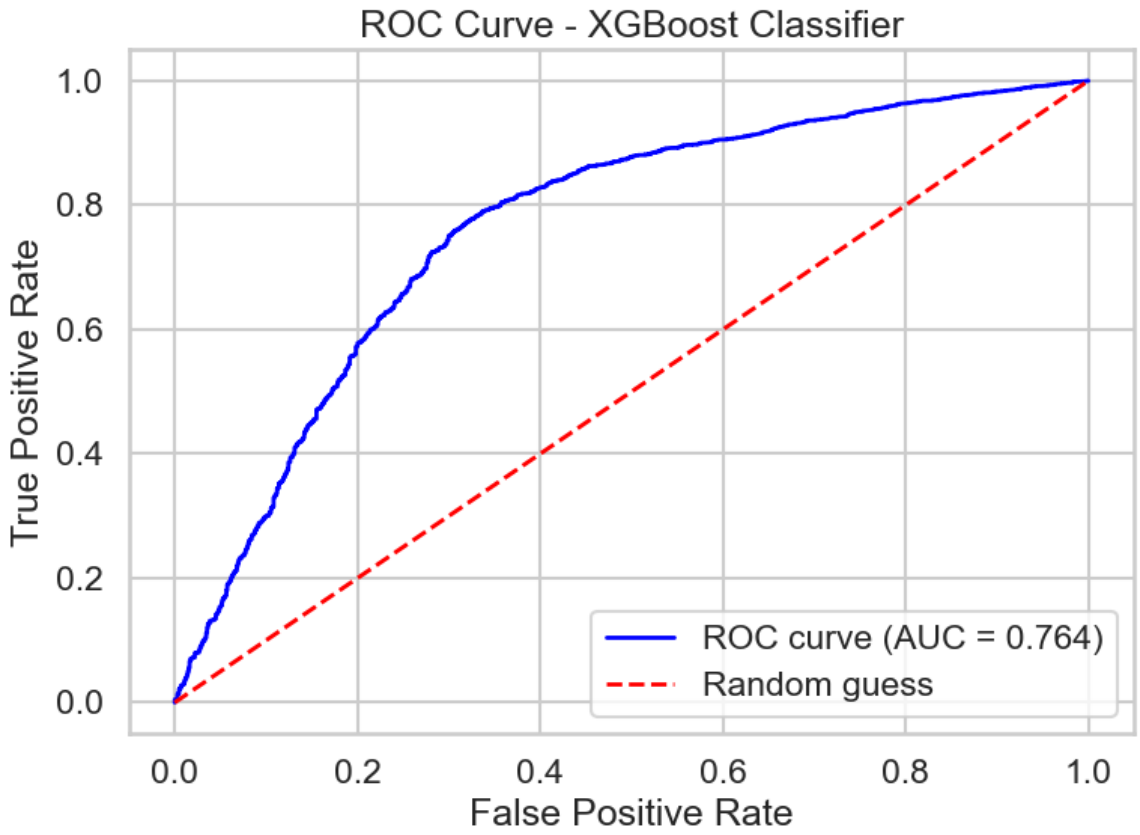

Figure: ROC Curbe for XG Boost Classifier.png

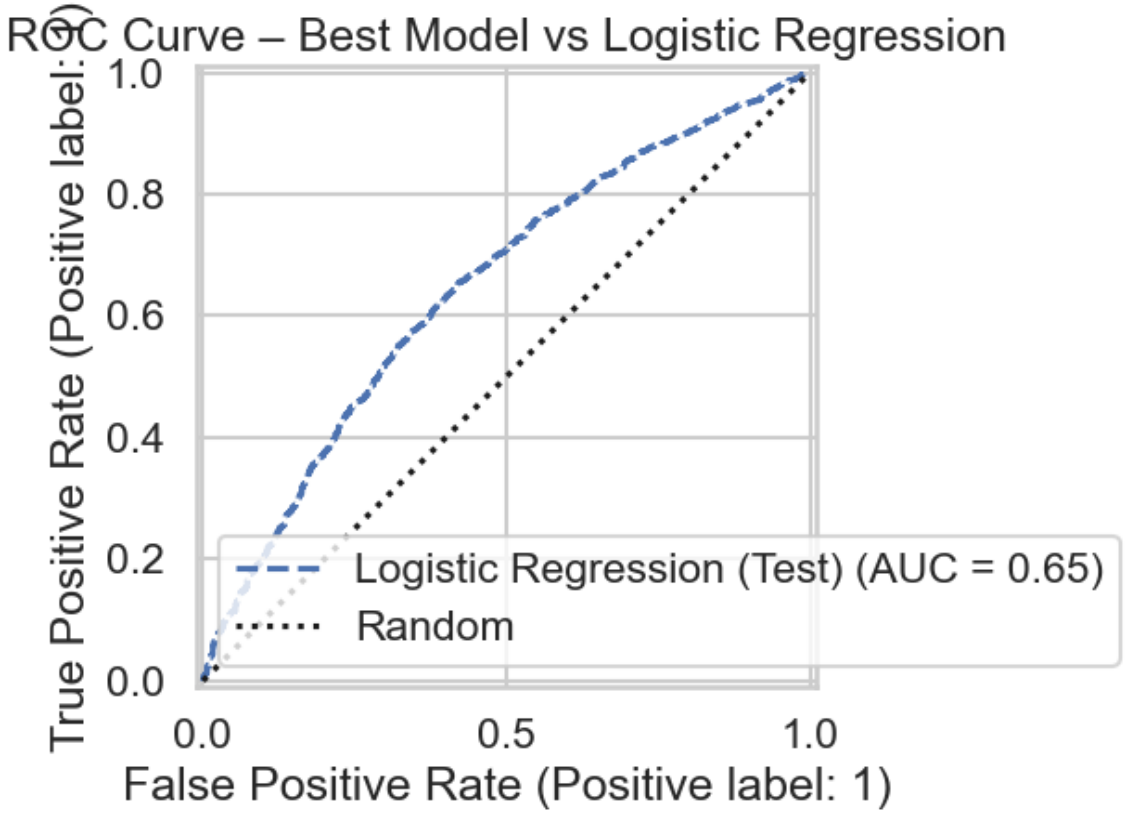

Figure: Random Forest vs Logistic Regression AUC.png

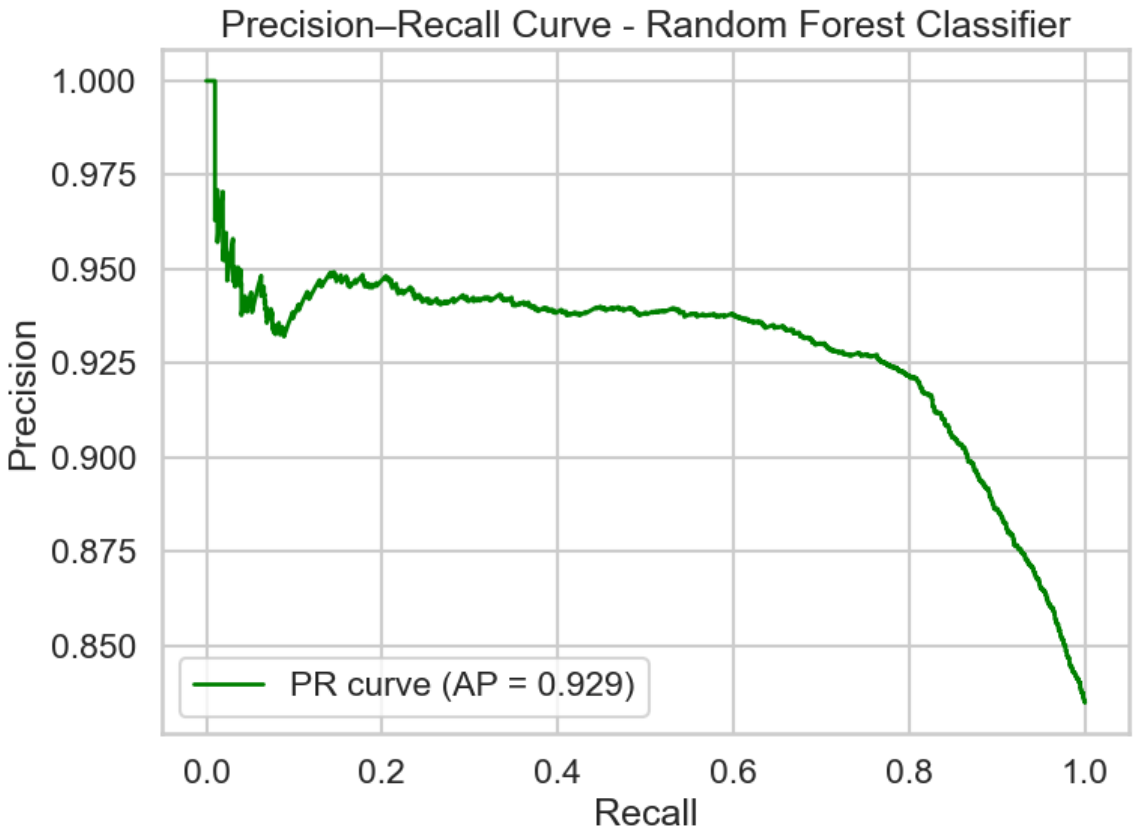

Figure: Random Forest Recall Curve.png

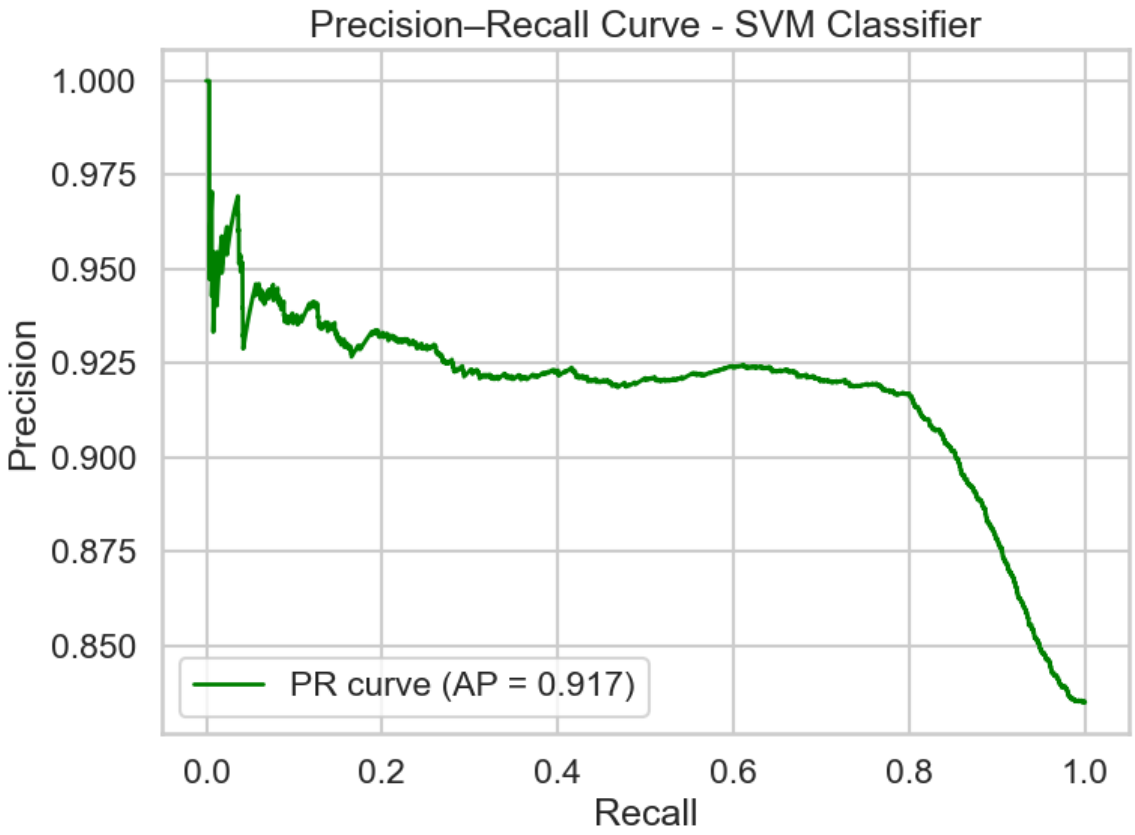

Figure: Recall curve SVM.png

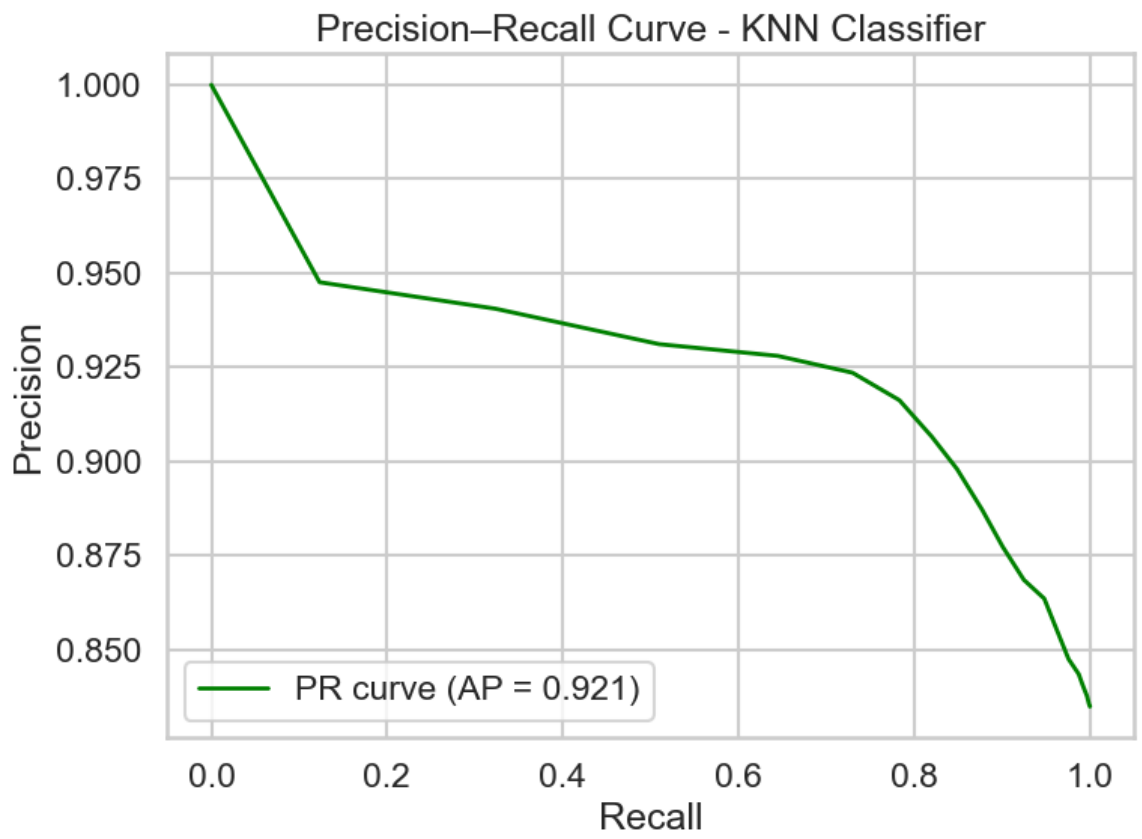

Figure: Recall KNN Classifier.png

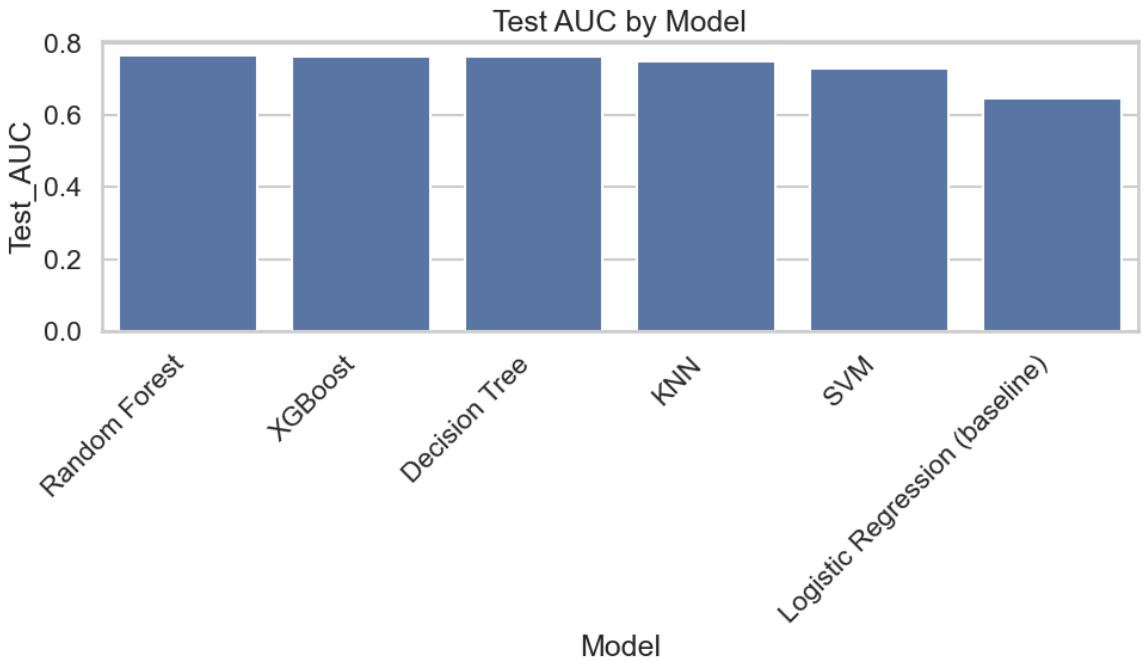

Figure: Test AUC by Model.png

Figure: Test F1 score by Model.png

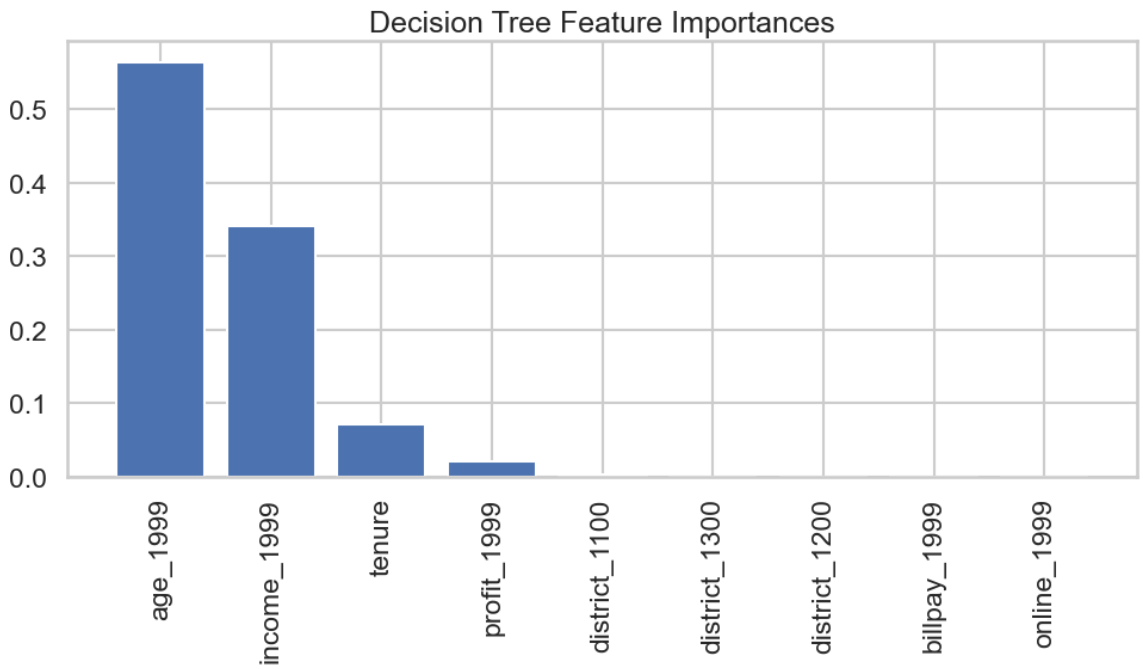

Figure: Decision Tree Feature Importance.png

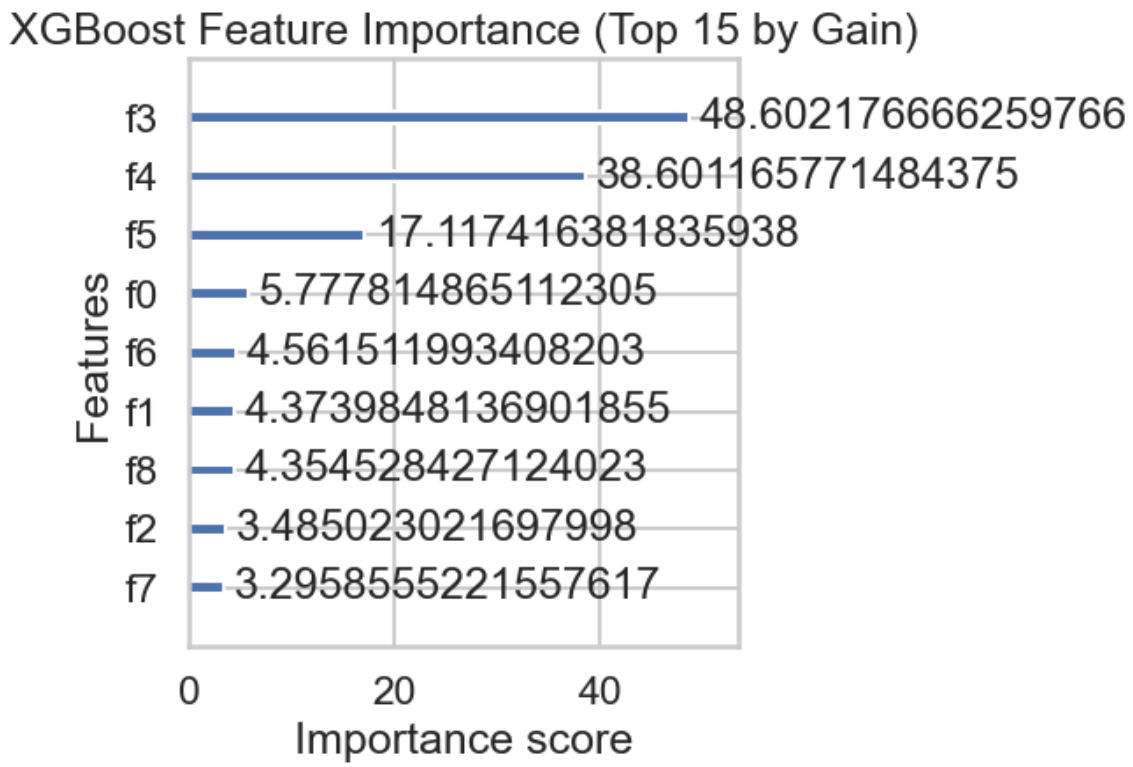

Figure: Feature Importance for XGBoost.png

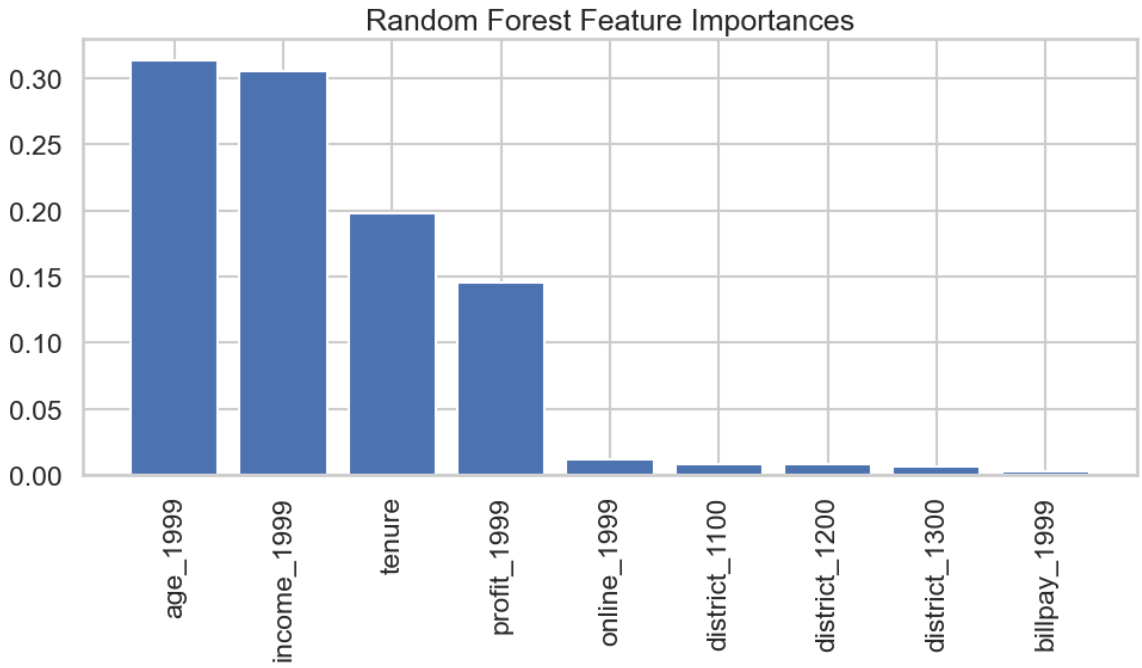

Figure: Random Forest Feature Importance.png

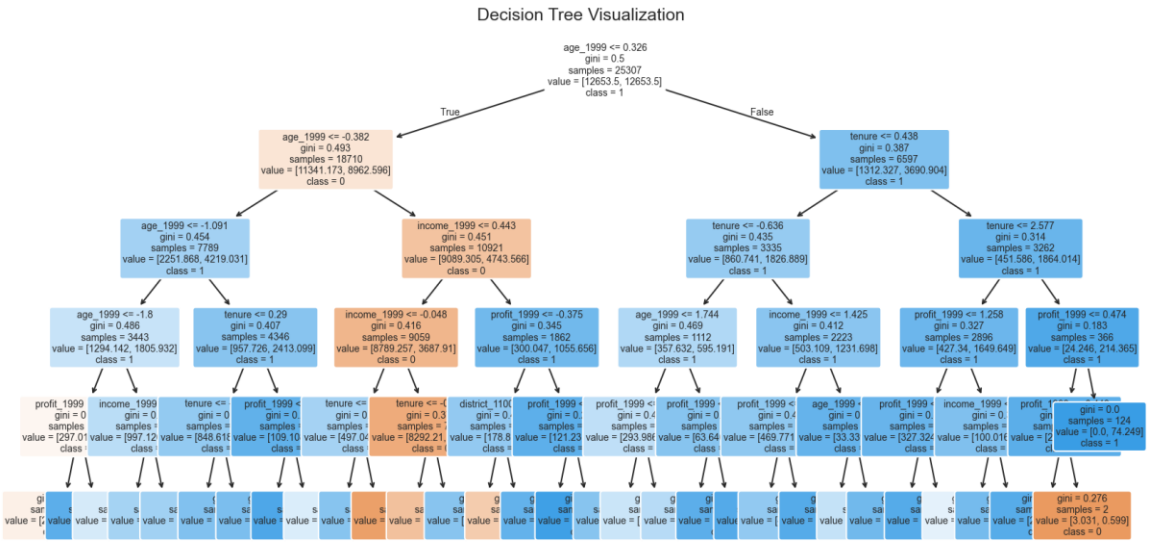

Figure: Decision Tree Structure.png

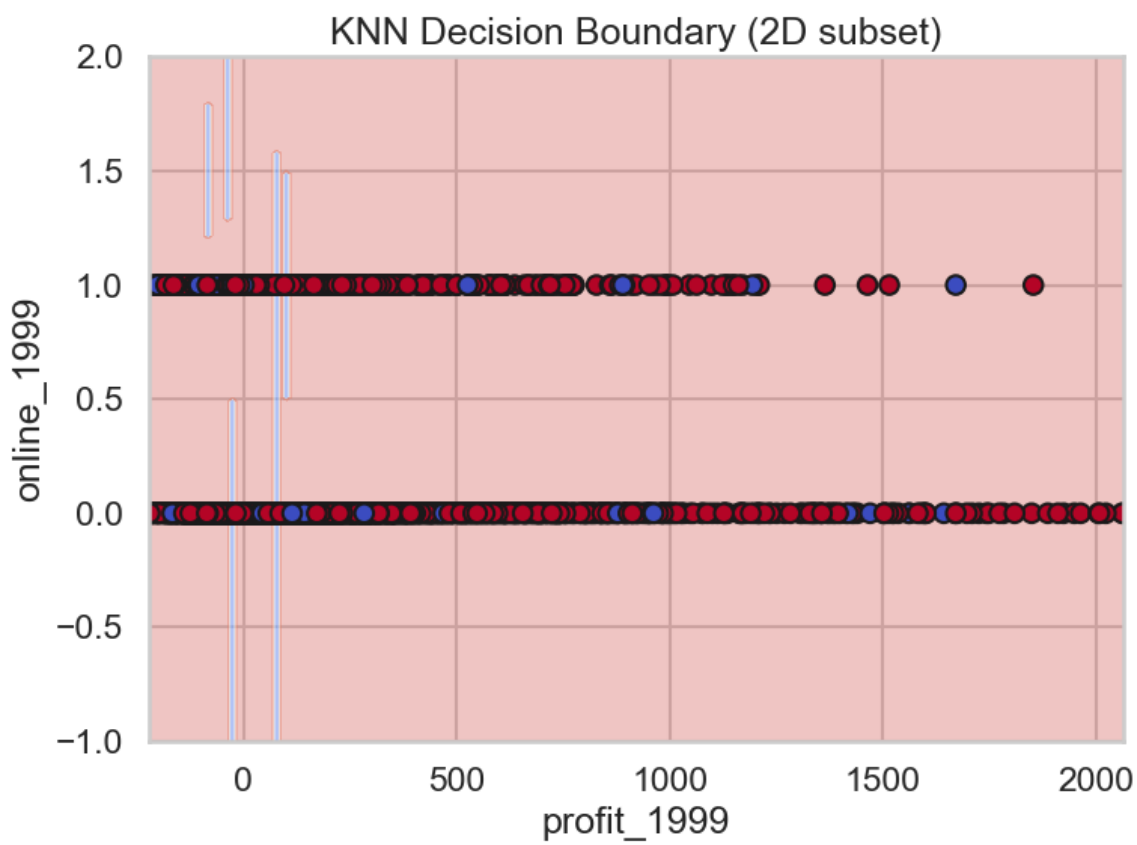

Figure: KNN Decision Boundary.png

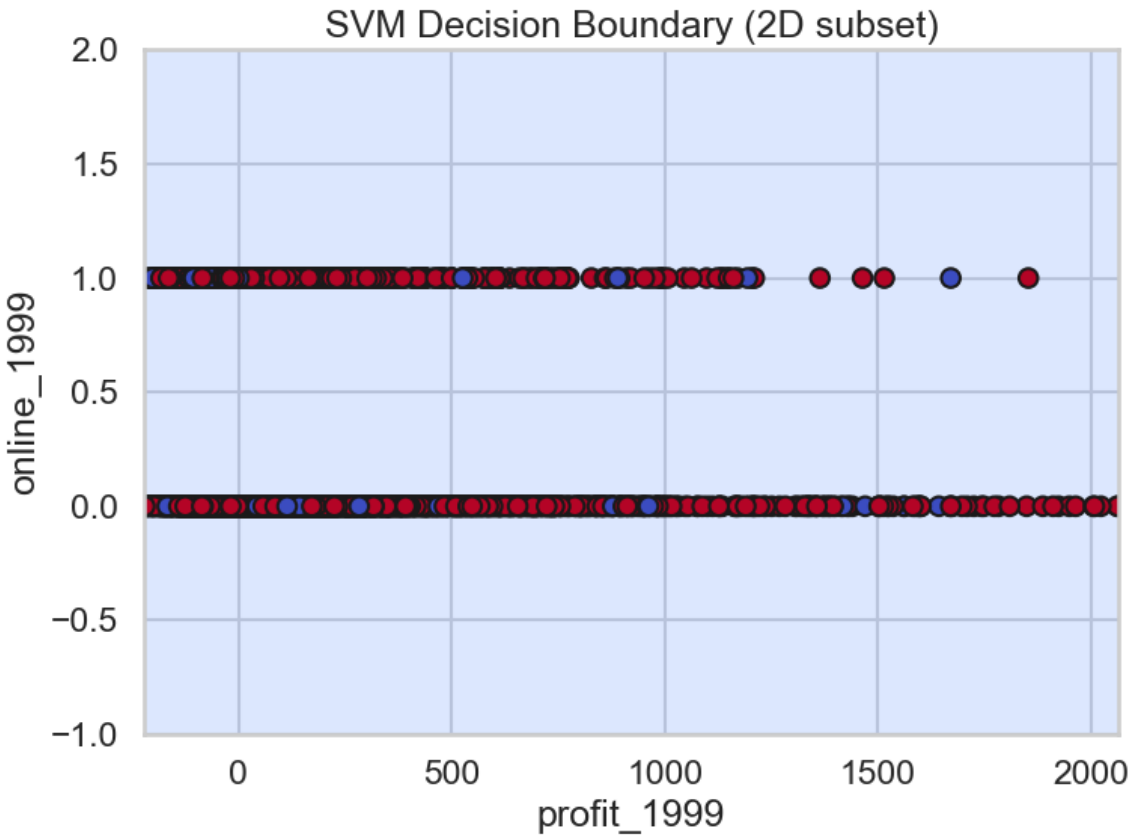

Figure: SVM Decission Boundary.png